Sign-Up

A native Android registration form with live validation, a dynamically enabled submit button, and a local account confirmation screen.

Assignment coverage

Required fields - Name, email, password and confirmation use XML inputs. Name: 2-80 characters. Password: 8-128 characters with at least one letter and one digit.

Dynamic validation - TextWatcher updates the submit state. Only valid fields and matching passwords enable Create account. Invalid fields show feedback on focus loss.

Local registration - App-private storage rejects duplicate emails without case sensitivity. Salted PBKDF2-HMAC-SHA256 derivation runs off the UI thread; the pending task survives Activity recreation.

Run and review

Open the repository root in Android Studio. Install SDK 35 and Build Tools 35.0.0; use JDK 17 or 21. Sync Gradle, then run the signup module on Android 8.0 / API 26 or later.

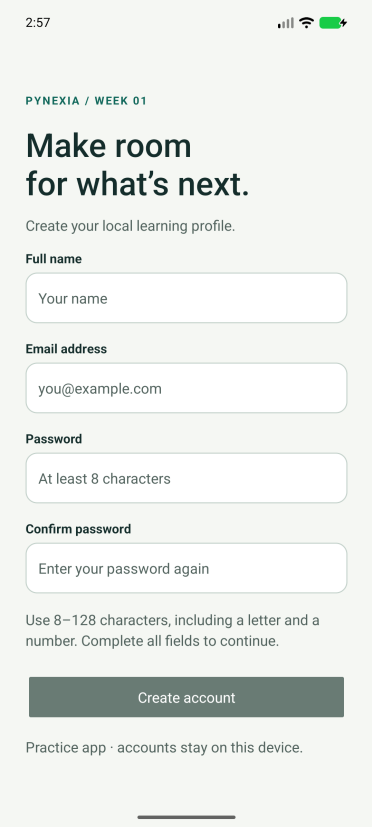
1. Try an invalid email and mismatched passwords; Create account stays disabled.
2. Use name Alex Learner, email alex@example.com, password Practice123 and matching confirmation.
3. Tap Create account and confirm the profile confirmation screen appears.
4. Return and reuse the same email to check duplicate handling. Clear app data to reset local accounts.

Scope

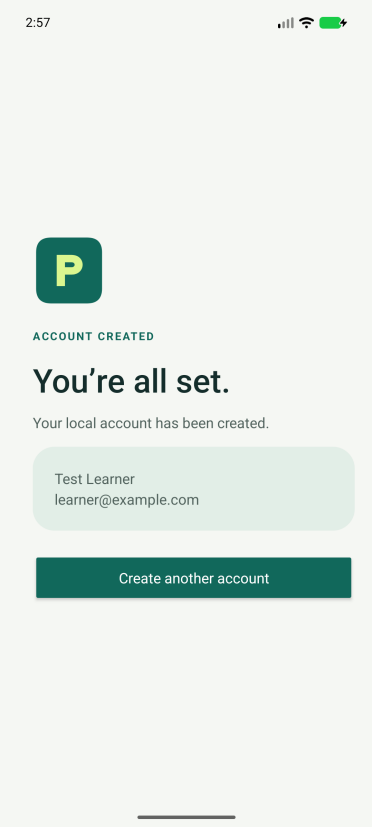
Local-only learning exercise: no backend, email verification, recovery or shared login account. Passwords are not saved across screen recreation. Autofill is disabled for this practice form.

Actual app screens

Captured from the tested native Android application, not browser mockups.



Registration form



Local account confirmation

Executed checks

JVM tests passed: 3 | Device tests passed: 2 | Lint errors: 0 | Lint warnings: 7

Three JVM tests cover validation rules. Two device tests cover submit states, local account creation without plaintext password storage, duplicate emails and registration across Activity recreation.

Debug APK built and signature verified. Tests ran on a Pixel 8 emulator, Android 17 / API 37 (37.1 image), x86_64. Zero test failures, errors or skips. No physical-device or API 26 execution. Lint warnings are retained without suppression; inspect verification/ for the findings.

Reproduce

.\gradlew.bat :signup:assembleDebug :signup:testDebugUnitTest :signup:lintDebug

.\gradlew.bat :signup:connectedDebugAndroidTest

The second command requires a running emulator or connected device. APKs are debug-signed evaluation builds.

Reference: https://developer.android.com/develop/ui/views/touch-and-input

Explanation & submission

يتحقق التمرين من الاسم والبريد الإلكتروني وكلمة المرور وتأكيدها أثناء الكتابة. لا يتفعل زر إنشاء الحساب إلا عند صحة جميع الحقول. تحفظ الحسابات داخل التطبيق محليا، وترفض عناوين البريد المكررة. تشتق كلمات المرور مع ملح عشوائي بدل حفظها كنص واضح، وتنفذ العملية خارج مسار واجهة المستخدم.

للتجربة: أدخل بيانات صحيحة وكلمة مرور من ثمانية أحرف على الأقل تتضمن حرفا ورقما، ثم كرر البريد نفسه للتحقق من منع التكرار. هذا التطبيق مستقل عن تمرين تسجيل الدخول ولا يرسل البيانات إلى خادم.

Repository URL

https://github.com/ENGEyad/pynexia-week1-signup

Live Demo / Hosted URL

https://engeyad.github.io/pynexia-week1-signup/

Hosted project showcase with real screenshots and APK/PDF downloads. It is not an in-browser Android emulator.

Attach File / Archive

Sign-Up-App.zip - includes source, APK, screenshots, verification and this report; below the 10 MB limit. PDF may also be attached separately. APK is inside the ZIP because the supplied form does not list APK as a direct attachment type.

Notes / Implementation Remarks (1468 / 2000 characters)

Native Android Sign-Up App built with Java and XML. The form includes Name, Email, Password and Confirm Password. Field validation checks name length, email format, password rules and matching confirmation. Create account is enabled only when all fields are valid. Successful registration opens a confirmation Activity using an Intent.

Accounts are stored locally in app-private storage; duplicate emails are rejected case-insensitively. Passwords are salted and derived with PBKDF2-HMAC-SHA256, never stored as plaintext. Derivation runs off the UI thread, and pending registration survives Activity recreation. This assignment is local-only: there is no server or email verification, and these accounts are separate from the Simple Login App's demo credentials.

Run: open the source folder in Android Studio, sync Gradle, and run signup on Android 8.0/API 26 or later. Use a 2-80 character name, a valid email, and an 8-128 character password containing a letter and a digit; enter the same password in both fields.

Validation: debug APK built successfully; 3 JVM tests and 2 instrumentation tests passed on a Pixel 8 emulator (Android 17/API 37). Lint completed with zero errors; warnings are documented. Source, Gradle Wrapper, APK, screenshots and verification reports are included in the attached ZIP.

Hosted URL: project showcase with actual screenshots and APK/PDF downloads; not a browser emulator. The ZIP also includes a bilingual implementation report.